

Nexivarin 10 mg Tablets — eCTD Submission
Module 2 — CTD Summaries

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PharmaCorp Inc.
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2.3 Quality Overall Summary (QOS)

2.3.S Drug Substance

2.3.S.1 General Information

The drug substance is nexivarin hydrochloride, a dihydropyridine calcium channel blocker. The proposed INN is 'nexivarin'. Detailed nomenclature, structural formula, and physicochemical properties are provided in **Section 3.2.S.1** of Module 3 (refer to Table 1 in Section 3.2.S.1 for the complete physicochemical property summary).

Property	Value
Molecular Formula	C ₁₈ H ₁₈ N ₂ O ₂ · HCl
Molecular Weight	479.35 g/mol
CAS Number	127620-12-7 (hypothetical)
Physical Form	White to off-white crystalline powder
Melting Point	196–200 °C (decomposition)
Solubility (water, 25°C)	2.8 mg/mL (slightly soluble)
Solubility (0.1N HCl)	18.4 mg/mL (freely soluble)
log P	3.05
pKa	9.4 (basic amine)
BCS Classification	Class II (low solubility, high permeability)

Table 2.3.S.1-1 above summarizes key physicochemical properties; see Section 3.2.S.1.3 of Module 3 for the complete general properties data, including solubility across pH range, polymorphic characterization, and hygroscopicity data.

2.3.S.2 Manufacture

Nexivarin hydrochloride is manufactured by PharmaCorp API Division at the Newark, NJ facility (DMF #32456). The synthesis involves a convergent 5-step route. Critical process parameters (CPPs) have been identified and are controlled within validated ranges. The detailed manufacturing process flow diagram, process description, and batch records are provided in **Section 3.2.S.2** of Module 3.

Refer to Section 3.2.S.2.2 (Module 3) for the complete synthesis scheme, including the process flow diagram (Figure 3.2.S.2-1) and a description of each synthetic step with critical process parameters and in-process controls. Manufacturer details are provided in Section 3.2.S.2.1.

2.3.S.3 Characterisation

The structure of nexivarin hydrochloride has been confirmed by a comprehensive suite of spectroscopic and analytical methods including ¹H-NMR, ¹³C-NMR, COSY, HMBC, IR, UV-Vis, ESI-MS, HRMS, XRPD, and elemental analysis. Full characterisation data and spectra are provided in **Section 3.2.S.3** of Module 3.

Refer to Section 3.2.S.3 (Module 3) for the complete structural elucidation package, including annotated NMR spectra (Figures 3.2.S.3-1 through 3.2.S.3-5), IR spectrum (Figure 3.2.S.3-6), and XRPD diffractogram confirming Form I polymorph (Figure 3.2.S.3-7).

2.3.S.4 Control of Drug Substance

The drug substance specification and analytical methods are described below. The complete specification, method descriptions, validation reports, and batch analysis data are provided in **Section 3.2.S.4** of Module 3. See Table 2.3.S.4-1 for the specification summary.

Test	Specification	Method
Appearance	White to off-white powder	Visual
Identification (IR)	Conforms to reference standard	In-house IM-01
Identification (HPLC)	RT matches reference $\pm 2\%$	In-house IM-02
Assay (HPLC)	98.0 – 102.0% (anhydrous)	In-house AM-01
Related Substances (any single / total)	Any single $\leq 0.15\%$; Total $\leq 0.50\%$	In-house AM-02
Water Content (KF)	$\leq 0.5\%$	USP <921>
Residue on Ignition	$\leq 0.1\%$	USP <281>
Heavy Metals	≤ 10 ppm (as Pb)	ICH Q3D / ICP-MS
Particle Size (d90)	≤ 100 μm	Laser diffraction
Polymorphic Form	Form I	XRPD
Microbial Limits	TAMC ≤ 1000 cfu/g; TYMC ≤ 100 cfu/g	USP <61>/<62>

As listed in Table 2.3.S.4-1 above, the drug substance specification encompasses all ICH Q6A mandatory tests. The full specification and analytical method validation data are provided in Sections 3.2.S.4.1 and 3.2.S.4.3 of Module 3, respectively. Batch analysis results for validation and clinical batches are listed in Section 3.2.S.4.4 (refer to Table 1 in Section 3.2.S.4.4 for batch analysis summary).

2.3.S.7 Stability

Stability studies on nexivarin hydrochloride were conducted per ICH Q1A(R2) and Q1E guidelines. Long-term data (25°C/60% RH) are available for 36 months; accelerated data (40°C/75% RH) are available for 6 months. All results meet specifications. The stability design and full data tables are provided in **Section 3.2.S.7** of Module 3.

Refer to Section 3.2.S.7.3 (Module 3) for the complete drug substance stability data tables (see Table 1 in Section 3.2.S.7.3 for long-term and accelerated stability results, and Figure 3.2.S.7-1 for the assay trend plot). The proposed re-test period of 36 months is supported by the data provided in that section.

2.3.P Drug Product

2.3.P.1 Description and Composition

Nexivarin 10 mg Tablets are formulated as round, biconvex, film-coated tablets containing 10 mg nexivarin hydrochloride per tablet. The complete formulation composition and excipient function are provided in **Section 3.2.P.1**. See Table 2.3.P.1-1 below for the composition summary.

Component	Function	Amount/Tablet (mg)	% w/w
Nexivarin Hydrochloride	Active	10.00	5.56
Microcrystalline Cellulose PH 102	Filler/Binder	80.00	44.44
Lactose Monohydrate	Filler	55.00	30.56
Croscarmellose Sodium	Disintegrant	8.00	4.44
Colloidal Silicon Dioxide	Glidant	2.00	1.11
Magnesium Stearate	Lubricant	2.00	1.11
Opadry White 03F280023	Film coat	13.00	7.22
Purified Water*	Film coat vehicle	q.s.	—
Film-coated tablet weight		180.00	100.00

*As depicted in Table 2.3.P.1-1 above, the core tablet contains six excipients. Excipient selection rationale, compatibility data, and compendial status are detailed in Section 3.2.P.2.3 (Module 3). *Purified water is removed during film-coating processing and is not present in the finished product.*

2.3.P.2 Pharmaceutical Development

Formulation development followed ICH Q8(R2) Quality by Design principles. The Quality Target Product Profile (QTPP), Critical Quality Attributes (CQAs), and the design space are described in **Section 3.2.P.2** of Module 3. The pharmaceutical development report is provided in Section 3.2.P.2.

Refer to Section 3.2.P.2.4 (Module 3) for the formulation optimization data, including the DoE results and design space (see Figure 3.2.P.2-1 for the Plackett-Burman screening results and Figure 3.2.P.2-2 for the central composite design contour plots). The final formulation was selected based on dissolution performance data listed in Table 1 of Section 3.2.P.2.4.

2.3.P.3 Manufacture

Nexivarin 10 mg Tablets are manufactured by PharmaCorp Manufacturing LLC, Newark, NJ via wet granulation (high-shear granulation), fluid bed drying, milling, blending, tablet compression, and film coating. The manufacturing process description, process flow diagram, and in-process controls are provided in **Section 3.2.P.3** of Module 3. The batch formula is provided in Section 3.2.P.3.2 (refer to Table 1 in Section 3.2.P.3.2 for the commercial batch formula).

The complete manufacturing process step descriptions and key process parameters are listed in Table 1 of Section 3.2.P.3.3 (Module 3). The process validation data, including blend uniformity, dissolution, and content uniformity results, are provided in Section 3.2.P.3.5. See Section 3.2.P.3.3 for the process flow diagram (Figure 3.2.P.3-1).

2.3.P.5 Control of Drug Product

The drug product specification encompasses all ICH Q6A mandatory tests. See Table 2.3.P.5-1 for the specification summary. The complete specification with acceptance criteria, analytical methods, and validation data are provided in **Section 3.2.P.5** of Module 3.

Test	Specification	Method
Appearance	White to off-white round film-coated tablet	Visual
Identification	Positive by HPLC UV	In-house PM-01
Assay	95.0 – 105.0% label claim	In-house PM-02 (HPLC)
Uniformity of Dosage Units	AV ≤ 15.0 (USP <905>)	USP <905>
Dissolution	≥ 80% in 30 min (USP App. II, 75 rpm, pH 6.8 PBS)	USP <711>
Related Substances	Any single ≤ 0.2%; Total ≤ 0.5%	In-house PM-03
Water Content	≤ 3.0%	USP <921>
Hardness	8 – 18 kP	Pharmacopoeial
Friability	≤ 1.0%	USP <1216>
Microbial Limits	TAMC ≤ 1000 cfu/g; TYMC ≤ 100 cfu/g	USP <61>/<62>

As listed in Table 2.3.P.5-1, the drug product specification includes all critical quality attributes. Analytical method validation reports for PM-01 through PM-03 are provided in Section 3.2.P.5.3 (Module 3). Batch analysis results for clinical and commercial-scale batches are provided in Section 3.2.P.5.4 (data listed in Table 2 of Section 3.2.P.5.4).

2.3.P.8 Stability

Drug product stability studies were conducted per ICH Q1A(R2). Long-term data (25°C/60% RH) are available for 24 months; accelerated data (40°C/75% RH) for 6 months. All results are within specifications. The complete stability protocol, data tables, and trend analyses are provided in **Section 3.2.P.8** of Module 3.

Refer to Section 3.2.P.8.3 (Module 3) for the complete drug product stability data tables (see Table 1 in Section 3.2.P.8.3 for long-term data and Table 2 for accelerated data). The dissolution trend over time is depicted in Figure 3.2.P.8-1 in Section 3.2.P.8.3. The proposed shelf life of 24 months is fully supported by the stability data provided in Section 3.2.P.8.